GeoAI-Based Landslide and Slope-Failure Risk Mapping

A Per-Building Risk Assessment for the
Sidi Bou Said Hillside, Tunisia

Internship Report

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Author:

Zeineb Sinini

Manouba School of Engineering

Abstract

This report presents a GeoAI-based landslide and slope-failure risk mapping system developed for the hillside of Sidi Bou Said, Tunisia, based on a real, well-documented triggering event: Storm Harry (January 2026). In the absence of a formal regional landslide inventory, a label set of confirmed instability zones was constructed from first principles, combining Sentinel-2 NDVI change detection, slope-based terrain screening, and multi-date high-resolution visual verification against historical satellite imagery. Two zones, along Rue El Hadi Zarrouk and along the southwestern coastal slope, were independently confirmed as pre-existing, chronically unstable features significantly accelerated by the storm, corroborated by a municipal evacuation and road-closure notice. A hybrid static susceptibility model was built using both a Frequency Ratio analysis and a Random Forest classifier, cross-validated against each other; a sensitivity check isolated elevation as a partially confounding, spatially clustered artifact, leaving slope, terrain roughness (TRI), and aspect as the converged, defensible drivers of instability at this site. A rainfall-triggering threshold was calibrated from CHIRPS precipitation data against the storm event, and a single-pair Sentinel-1 InSAR coherence product was used as corroborating, low-weight evidence of ground disturbance. The susceptibility layer was then combined with building footprints (OpenStreetMap) and proximity to confirmed instability into a final per-building risk score, while footprint area was kept as a separate vulnerability proxy and the rainfall and InSAR layers served as supporting evidence rather than direct inputs to the score. Validation against the documented closure corridor shows that the model's highest-risk predictions concentrate in the area that was actually closed and partially evacuated during the real event, without the model having been informed of specific building locations. The final deliverable is a four-map series presenting terrain evidence, hazard, exposure and vulnerability, and final per-building risk for the Sidi Bou Said hillside.

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Chapter 1

Introduction

1.1 Motivation

Landslides and slope failure are a continual threat in hilly and coastal regions, where rain-induced instability can endanger structures with little warning. Mapping this hazard at the level of a specific building, rather than as a generalized susceptibility layer, is far more useful to stakeholders: it shows which buildings actually need monitoring or intervention, rather than simply stating that a slope is risky in general.

This project was carried out as part of a one-month internship at the Association of Geospatial Intelligence (AGI) in Tunis, under the topic "GeoAI-Based Landslide and Slope-Failure Risk Mapping for Hillside and Mountain Structures." The goal was to build a single per-building risk product by combining a static susceptibility model, dynamic triggers such as rainfall and ground deformation, and structural exposure and vulnerability.

1.2 Scoping Decisions

Implementing every subsystem in the original proposal at full depth would have taken an estimated nine to seventeen weeks, depending on whether InSAR processing was built from scratch. Given the one-month timeframe, the project was scoped as follows:

- **Static susceptibility** was treated as the core, most thoroughly developed component.
- **Rainfall triggering** was modeled more simply, but still grounded in the published antecedent-rainfall threshold approach.
- **Ground deformation** relied on a single Sentinel-1 interferometric pair, processed through an on-demand service, rather than a full multi-pair time series.
- **Exposure and vulnerability** used readily available proxies in place of occupancy or construction-quality data, which were not accessible.

1.3 Study Area

The Area of Interest (AOI) is the hillside of **Sidi Bou Said**, a coastal village near Tunis, Tunisia. The AOI was intentionally narrowed from the broad internship brief to this one hillside, and tied to a real, dated triggering event, described in Chapter 4.

Chapter 2

Background and Related Methodology

2.1 Landslide Susceptibility Mapping

Static landslide susceptibility mapping typically relates terrain covariates, such as slope, aspect, curvature, terrain roughness, and topographic wetness, to the locations of known landslides. Two families of method are commonly used: knowledge-based statistical methods such as Frequency Ratio (FR), which suit data-poor regions well since they need no large training set and remain easy to interpret; and data-driven methods such as Random Forest (RF) or Gradient Boosting, which can model nonlinear relationships between covariates but need a larger, more reliable label set to avoid overfitting.

2.2 The Hazard, Exposure, and Vulnerability Framework

Disaster risk is conventionally expressed as

$$\text{Risk} = \text{Hazard} \times \text{Exposure} \times \text{Vulnerability}, \tag{2.1}$$

where Hazard is the physical likelihood and strength of the event, Exposure is what sits in harm's way, and Vulnerability is how susceptible that exposed asset is to damage. In this project, Hazard is represented by the static susceptibility surface, with the rainfall trigger and InSAR coherence used as supporting evidence; Exposure is the building footprints; and Vulnerability is represented by the distance from a confirmed instability zone, with footprint area kept as a separate proxy.

Chapter 3

Data Sources

The project drew on seven data sources in total, summarized in Table 3.1, spanning satellite imagery, a digital elevation model, rainfall data, building footprints, and the municipal notice that the whole study is based on.

Table 3.1: Data sources used in this project.

Data	**Source**	**Use**
Sentinel-2 L2A	Google Earth Engine	NDVI change detection (before and after)
Sentinel-1 SLC	Alaska Satellite Facility (ASF)	Single-pair InSAR coherence
Copernicus DEM GLO-30	Google Earth Engine	Terrain feature derivation
CHIRPS daily precipitation	Google Earth Engine	Rainfall antecedent-threshold analysis
High-resolution historical imagery	ESRI World Imagery Wayback	Multi-date visual verification
Building footprints	OpenStreetMap (via `osmnx`)	Exposure layer
Street network and closure corridor	OpenStreetMap (QuickOSM / Overpass)	AOI construction
Municipal evacuation and closure notice	Local press and municipal announcement	Label-set grounding, validation

These sources are each revisited in the methodology chapter that follows, where they are tied to the specific step of the pipeline they support.

Chapter 4

The Key Event: Storm Harry

Research into the study area uncovered a real, well-documented slope failure caused by **Storm Harry**, which struck Tunisia in late January 2026. Regional press reported it as the heaviest rainfall the region had seen in over 70 years, with nearly 300 mm falling in 48 hours. The event caused fatalities, evacuations, and a government emergency session.

A municipal notice in Arabic ordered the temporary evacuation of **eight properties** following civil protection inspections, and closed parts of **Rue El Hadi Zarrouk** at its intersection with **Avenue John Kennedy**, along with a section of Avenue John Kennedy up to a guesthouse called "Villa Didon." A geomorphologist noted that the hillside was already unstable, and that the storm acted as a trigger on pre-existing conditions rather than creating new instability outright. This fits with the site's clay-rich soils, which lose cohesion when saturated, and which are compounded by coastal erosion and decades of denser construction.

This event became the central reference point of the project. Rather than relying on Tunisia's sparse, effectively nonexistent landslide inventory, an evidence-based label set was built directly from this documented event.

Chapter 5

Methodology

5.1 Overview

Figure 5.1 summarizes the full pipeline, from AOI construction through to the final per-building risk map; each stage is described in detail in the sections that follow.

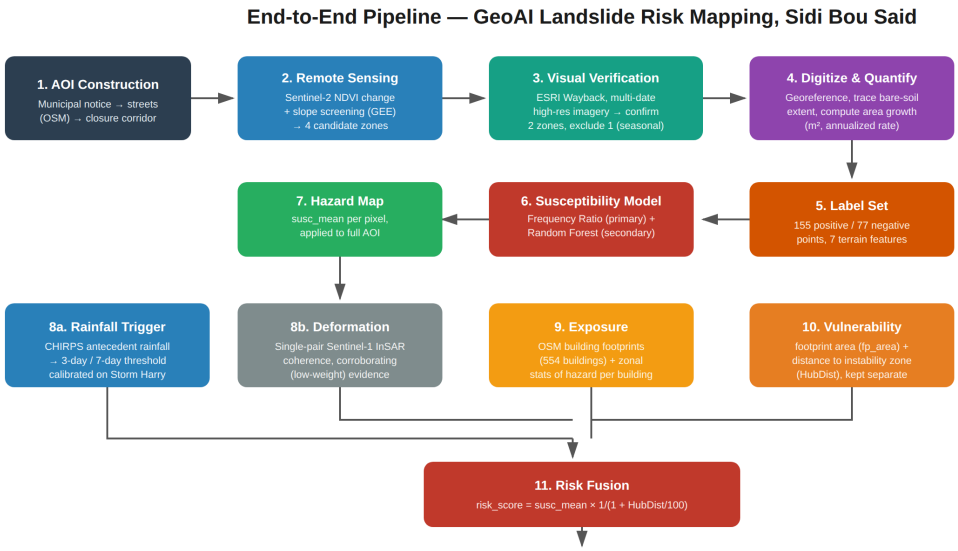


Figure 5.1: End-to-end pipeline overview, from AOI construction to final risk fusion.

Each box in this diagram corresponds to one of the methodology sections below, in the same left-to-right, top-to-bottom order.

5.2 AOI Construction

The closure-zone AOI was built directly from the municipal notice described in Chapter 4. The two named streets and their intersection were located through QuickOSM and Overpass Turbo queries against OpenStreetMap. An unnamed connecting road segment, not discoverable by name-based search, was found by querying all `highway` features in the area and visually isolating the connecting way. These street segments were merged, trimmed, and buffered into a corridor polygon, then clipped to a manually corrected land and coastline polygon to remove

an initial overshoot into the sea. The result is a defensible, evidence-grounded AOI, traceable at every step back to the real municipal document, rather than an arbitrary boundary.

5.3 Remote Sensing and Candidate Zone Identification

Sentinel-2 Level-2A imagery was retrieved via Google Earth Engine for a pre-storm date (9 January 2026, cloud cover around 0.02%) and a post-storm date (6 February 2026, cloud cover around 4.6%), cloud-masked using the Scene Classification Layer. The NDVI difference between the two dates was computed and thresholded (difference below -0.2) to isolate strong candidate zones of vegetation loss or bare-soil exposure. Four candidate zones emerged: a zone along Rue El Hadi Zarrouk, a southwestern coastal cluster, a hilltop zone near the village center, and a flat, low-lying cluster.

These zones were cross-checked against slope, computed from the Copernicus GLO-30 DEM. The flat cluster was dropped as inconsistent with slope-failure mechanics, most likely unrelated land-cover change, while the Zarrouk and southwestern zones sat on genuinely steep terrain and the hilltop zone was only moderately steep.

5.4 Multi-Date Visual Verification

The three remaining candidate zones were inspected using ESRI World Imagery Wayback, which provides dated historical high-resolution satellite imagery. For the Zarrouk and southwestern zones, imagery from 2021 to 2025 baselines was compared against February 2026: both showed a smaller, pre-existing bare-soil scar that grew substantially larger and more defined by 2026, consistent with chronic instability significantly accelerated by Storm Harry. For the hilltop zone, a same-season baseline (November 2021) was chosen to control for seasonal vegetation cycling; it showed dense, undisturbed canopy, ruling out seasonal browning as the explanation, but the observed change was diffuse thinning without a clean, bounded scar. The hilltop zone was therefore treated as weak, inconclusive evidence and excluded from the positive label set.

5.5 Digitization and Quantification

The confirmed Zarrouk and southwestern zones were georeferenced in QGIS, using four ground-control points per image with a residual error typically under ten pixels, and reprojected to EPSG:32632 (WGS 84 / UTM Zone 32N). Bare-soil extent polygons were manually digitized at each available date; in both zones the extent turned out to be a series of disconnected polygons rather than one continuous shape, reflecting the genuinely patchy nature of the exposed ground.

Table 5.1 reports the resulting area figures at each zone.

Table 5.1: Quantified bare-soil extent growth at the two confirmed instability zones.

Zone	**Baseline date**	**Baseline (m^2)**	**2026 (m^2)**	**Growth**	**Annualized rate**
Zarrouk	Aug 2024	9,083	58,691	6.5×	∼33,000 m^2/yr
Southwest	Nov 2021	52,857	188,030	3.6×	∼32,000 m^2/yr

The two zones' baseline dates differ substantially, so the raw growth multiples are not directly comparable, yet both show a strikingly similar annualized growth rate. This is a more compelling and defensible finding than the raw totals alone, and it points to a shared underlying instability process at both locations.

5.6 Label-Set Construction

Random points were drawn from the confirmed 2026 bare-soil extents, using density-based sampling given how fragmented the polygons were: 59 points fell within the Zarrouk zone and 96 within the southwestern zone, for a total of **155 positive points** (`class = 1`). For the negative class, points were sampled across a wider area while excluding a 75 m buffer around the positive zones, yielding **77 negative points** (`class = 0`) after filtering. Figure 5.2 walks through this workflow from start to finish.

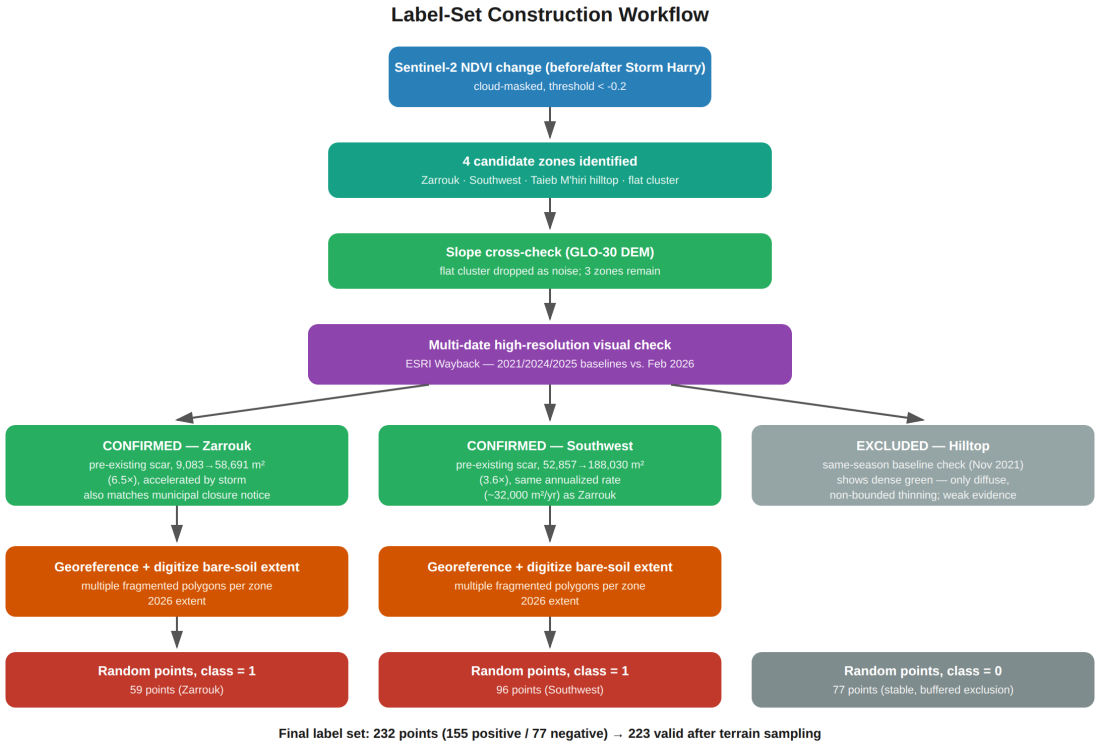


Figure 5.2: Label-set construction workflow, from NDVI detection through multi-date visual verification to the final positive and negative point set.

The resulting 232-point set is the foundation for everything that follows in the susceptibility modeling.

5.7 Terrain Feature Extraction

Seven terrain covariates were derived from the DEM in QGIS: elevation, slope, aspect, profile curvature, plan (tangential) curvature, Topographic Wetness Index (TWI, via GRASS `r.watershed`), and Terrain Ruggedness Index (TRI). An early mismatch, where the terrain rasters were clipped to a small AOI while the negative-label points extended across a much

wider area, caused a considerable loss of data. This was resolved iteratively, by re-scoping the DEM export extent and fixing a gap in the land and coastline polygon used for clipping. In the end, 151 of 155 positive points and 72 of 77 negative points kept complete, valid terrain data, for a final working set of **223 labeled points**.

5.8 Static Susceptibility Modeling

A hybrid approach combined a knowledge-driven and a data-driven method, illustrated in Figure 5.3.

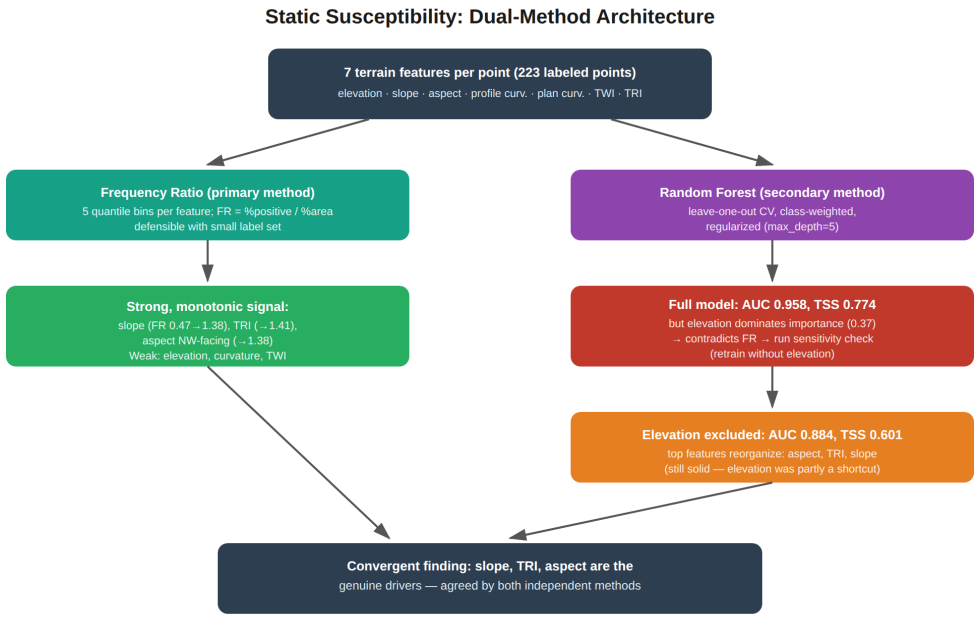


Figure 5.3: Dual-method susceptibility architecture: Frequency Ratio (primary) and Random Forest (secondary), with the elevation sensitivity check.

The two branches shown here, Frequency Ratio and Random Forest, are described in turn below, followed by the sensitivity check that reconciles a disagreement between them.

5.8.1 Frequency Ratio

Each terrain feature was split into five quantile-based bins. For each bin, the Frequency Ratio was computed as

$$\mathrm{FR} = \frac{\text{share of positive-class points in bin}}{\text{share of all points in bin}}, \tag{5.1}$$

where FR $>$ 1 signals that instability is over-represented in that terrain range, pointing to elevated susceptibility.

5.8.2 Random Forest

A Random Forest classifier was trained using leave-one-out cross-validation, a sensible choice given the small sample size. Class weighting handled the 151/72 class imbalance, and the model was regularized (maximum depth of five, minimum five leaf samples) to keep overfitting in check.

5.8.3 Elevation Sensitivity Check

The full model's feature importances ranked elevation far above every other covariate, directly contradicting the Frequency Ratio analysis, which had flagged elevation as the weakest and least consistent predictor. To investigate this, a sensitivity check retrained the Random Forest with elevation excluded; the result is discussed in Section 6.1.

5.9 Rainfall Trigger Analysis

Daily CHIRPS precipitation data were pulled for the AOI via Google Earth Engine, covering 1 November 2025 to 13 February 2026. Rolling antecedent-rainfall sums were computed over 1-, 3-, 7-, 15-, and 30-day windows, and the full record was ranked by cumulative rainfall to find the most extreme periods.

5.10 Ground Deformation

A single Sentinel-1 interferometric pair (Path 88, Frame 118, ascending, VV+VH) was processed through the Alaska Satellite Facility's on-demand InSAR service. The pair bracketed the storm cleanly: a reference scene from 17 January 2026, before the storm, and a secondary scene from 29 January 2026, after it, twelve days apart. The resulting coherence product was used to gauge ground disturbance, with one caveat worth stating plainly: coherence loss can also come from vegetation or land-cover change unrelated to ground movement.

5.11 Exposure and Vulnerability

Building footprints were retrieved from OpenStreetMap using `osmnx`, giving 554 footprints once an initially over-broad query extent was tightened. Each building's hazard value (`susc_mean`) came from zonal statistics against the Random Forest susceptibility raster. Two vulnerability proxies were then computed and deliberately kept as separate, transparent fields rather than collapsed into one composite score: footprint area (`fp_area`), and distance to the nearest confirmed instability zone (`HubDist`).

5.12 Risk Fusion

The final per-building risk score blends hazard and proximity-based exposure through a smooth distance-decay function:

$$\text{risk_score} = \text{susc_mean} \times \frac{1}{1 + \text{HubDist}/100}. \tag{5.2}$$

At zero distance from a confirmed zone, the multiplier equals 1, so the risk score simply equals the raw susceptibility. At 100 m it drops to 0.5, and at 300 m to 0.25. In short, a building only reaches a high final risk score if it scores highly on both dimensions: genuinely hazardous terrain and proximity to a place where instability has actually been observed, rather than on terrain steepness alone.

The rainfall trigger and the InSAR coherence do not enter this equation directly. The rainfall thresholds describe when the hillside is likely to fail, and the coherence product corroborates ground disturbance at the confirmed zones. Footprint area likewise remains a separate field rather than a term in the score.

Chapter 6

Case Study

6.1 Static Susceptibility

Table 6.1 summarizes the Frequency Ratio pattern for each terrain feature, from the strongest, most consistent signals to the weakest.

Table 6.1: Frequency Ratio results for the strongest and weakest terrain features.

Feature	**Pattern**	**Interpretation**
Slope	FR 0.47 (lowest) rising to 1.38 (steepest)	Strong, monotonic; textbook-consistent
TRI	FR up to 1.41 (roughest bin)	Strong, consistent with slope
Aspect	FR up to 1.38 (NW-facing, 285–359°)	Moderate, physically plausible
Elevation	Non-monotonic, falls to 0.23 at highest bin	Weak, unreliable
Curvature (both)	Near-flat, close to 1.0	Weak
TWI	Near-flat, close to 1.0	Weak

Table 6.2 shows the Random Forest's performance side by side with the elevation-excluded sensitivity check.

Table 6.2: Random Forest performance, full model versus elevation-excluded sensitivity check.

Metric	**Full model (7 features)**	**Elevation excluded (6 features)**
AUC-ROC	0.958	0.884
Sensitivity	0.940	n/a
Specificity	0.833	n/a
TSS	0.774	0.601
Top feature(s)	Elevation (0.366)	Aspect (0.295), TRI (0.252), Slope (0.238)

Once elevation was dropped, the top features reshuffled into aspect, TRI, and slope, lining up closely with the independent Frequency Ratio findings. This suggests elevation's dominant raw importance was partly a spatial-clustering artifact: the two confirmed zones happen to sit in a similar elevation band, so the model may have latched onto location rather than a genuine causal driver. Even so, the full model was kept for the final susceptibility map, since it still delivered the strongest measured accuracy.

Figure 6.1 shows the resulting pixel-wise susceptibility map, produced by applying the trained Random Forest model across the full terrain raster stack.

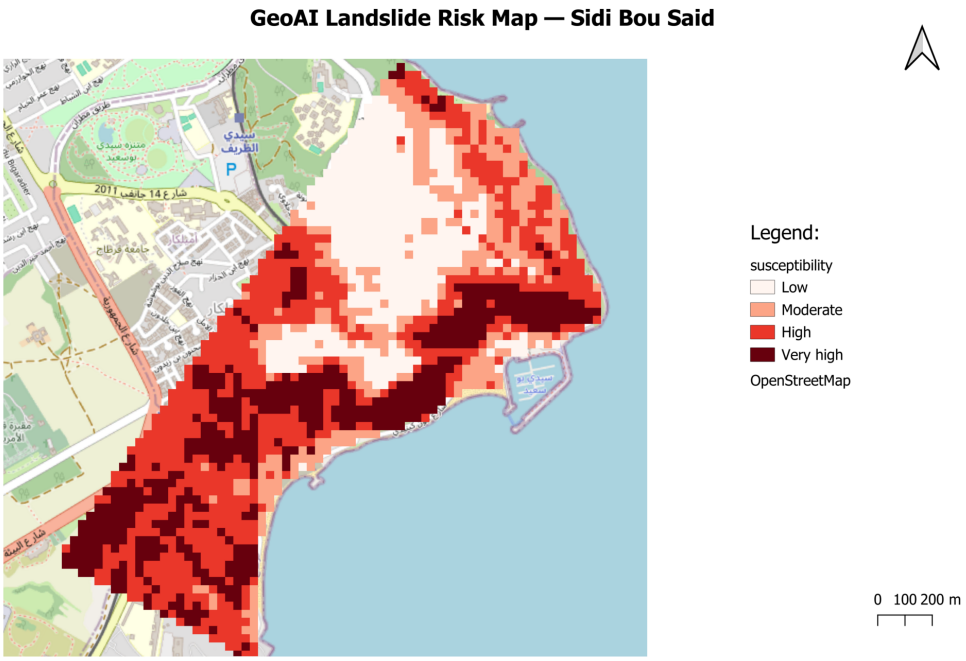


Figure 6.1: GeoAI landslide susceptibility map: Random Forest prediction across the Sidi Bou Said hillside.

The highest susceptibility concentrates along the coastal band, including the area around Rue El Hadi Zarrouk and the southwestern slope, where the two confirmed instability zones lie. Because the Random Forest can learn nonlinear interactions between features, the map shows a broad gradient of elevated risk rather than isolated hotspots, in contrast to the strictly multiplicative Frequency Ratio overlay, which is more restrained.

6.2 Terrain Covariate Maps

The five maps below show the individual terrain covariates that feed the susceptibility model, starting with slope and aspect and ending with the two curvature layers.

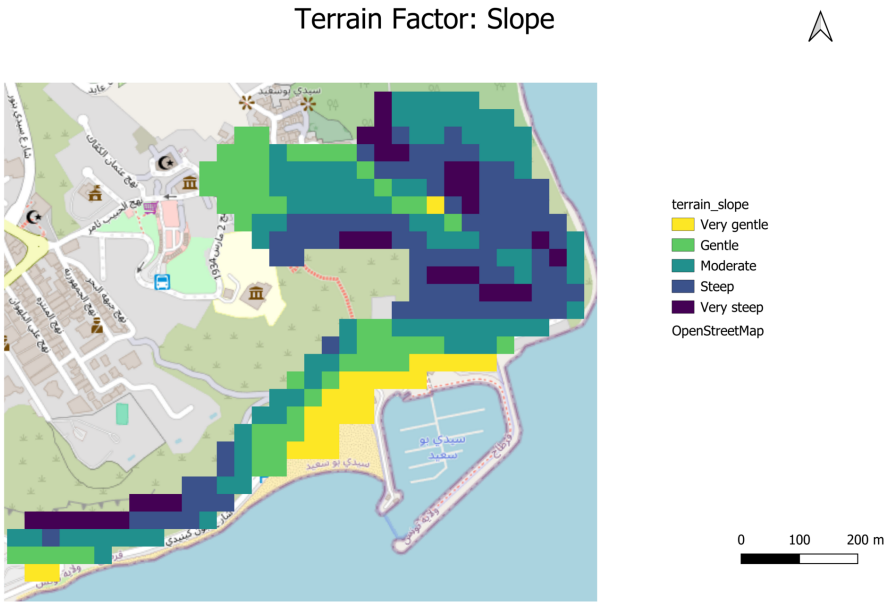


Figure 6.2: Slope map of the Sidi Bou Said AOI.

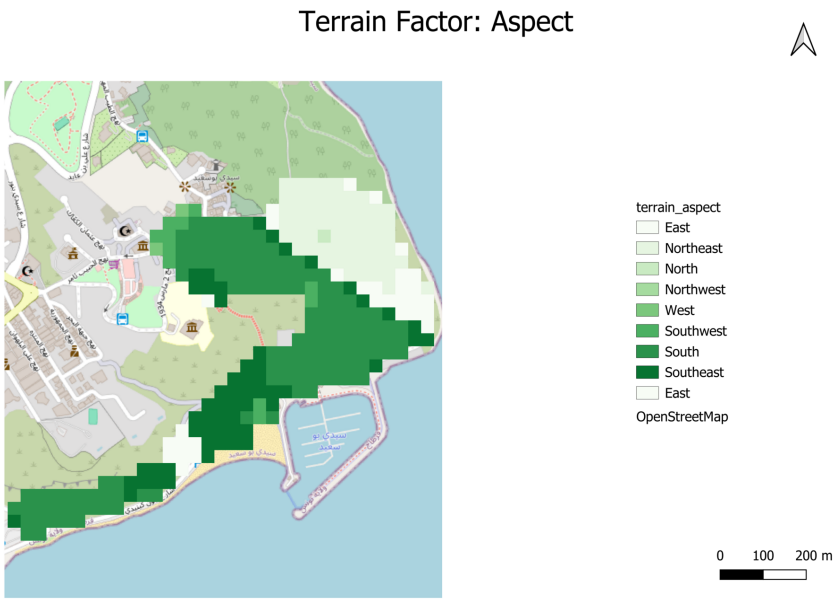


Figure 6.3: Aspect map of the Sidi Bou Said AOI.

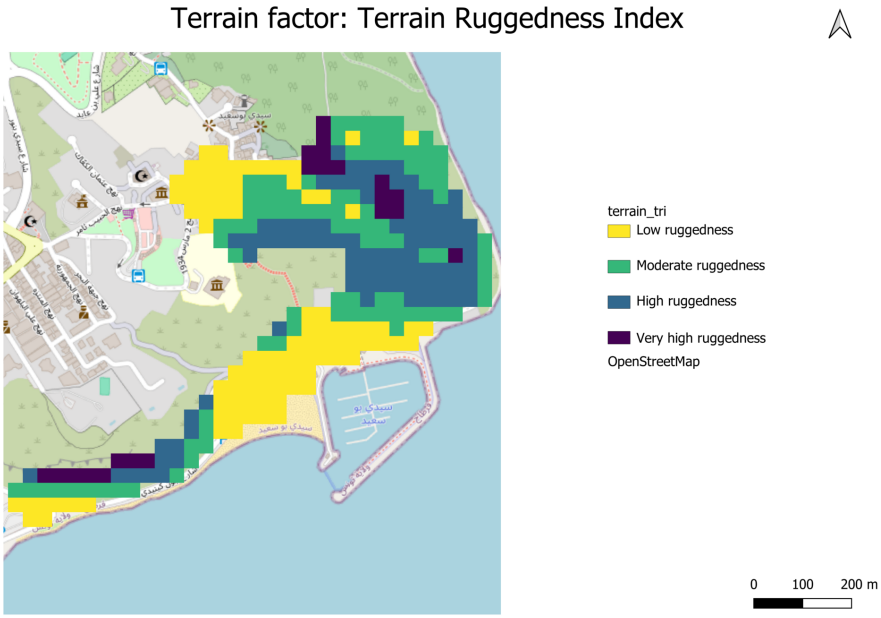


Figure 6.4: Terrain Ruggedness Index (TRI) map of the Sidi Bou Said AOI.

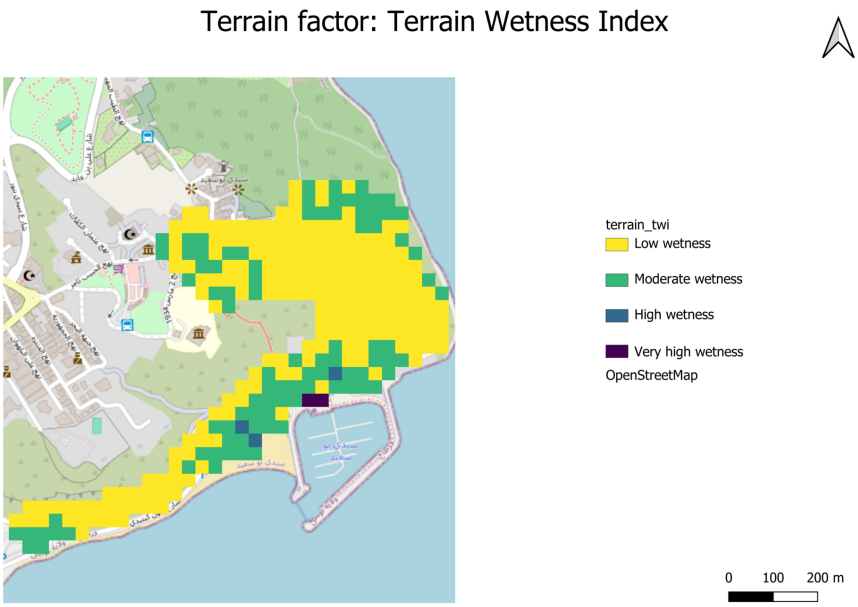


Figure 6.5: Topographic Wetness Index (TWI) map of the Sidi Bou Said AOI.

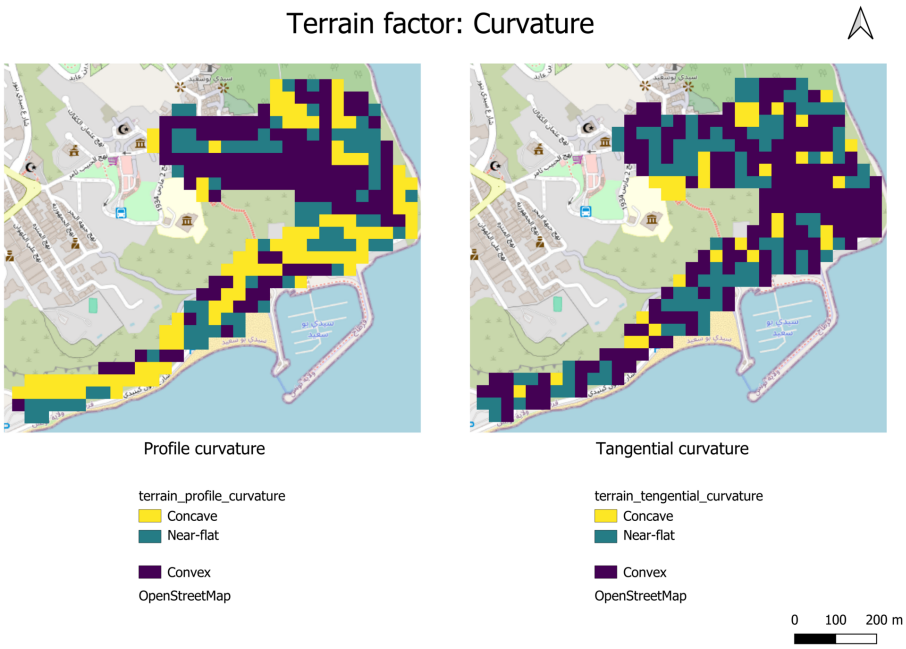


Figure 6.6: Curvature maps of the Sidi Bou Said AOI (left: profile curvature; right: tangential/plan curvature).

Together, these five layers, along with elevation, make up the seven covariates behind both the Frequency Ratio and Random Forest results above.

6.3 Rainfall Trigger

Figure 6.7 plots the daily rainfall record pulled from CHIRPS, which the antecedent-threshold analysis below is built on.

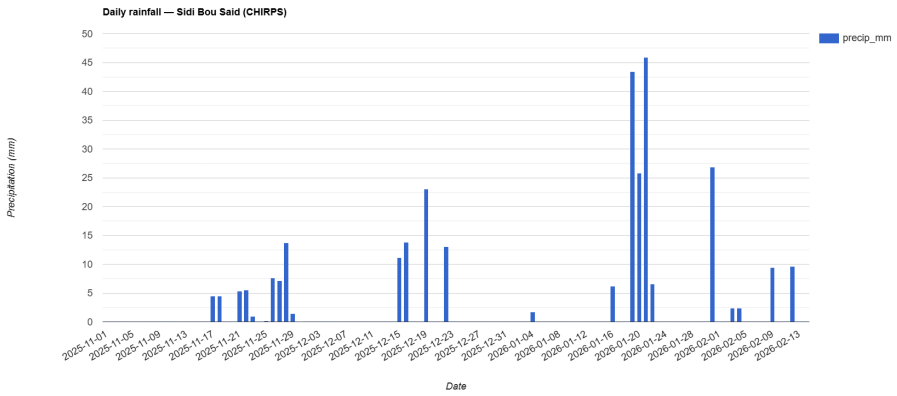


Figure 6.7: Daily rainfall at Sidi Bou Said (CHIRPS), 1 November 2025 to 13 February 2026, retrieved via Google Earth Engine.

Table 6.3 lists the resulting antecedent-rainfall thresholds at each time window.

Table 6.3: Calibrated antecedent-rainfall thresholds associated with Storm Harry.

Window	Peak cumulative rainfall	Date reached
3-day	115.1 mm	21 Jan 2026
7-day	127.9 mm	22–25 Jan 2026
15-day	148.5 mm	31 Jan 2026

For both the 3-day and 7-day windows, every one of the top five dates in the full ∼3.5-month CHIRPS record falls inside the documented Storm Harry period (19–25 January 2026). That is an unambiguous threshold-calibration result, even coming from a single event. One caveat is worth flagging: the CHIRPS-measured totals sit noticeably below the roughly 300 mm reported in press coverage. This gap is attributed to CHIRPS's known tendency to underestimate short, intense, localized rainfall at its native resolution of about 5.5 km. The relative timing and threshold calibration still hold, but the absolute magnitude is best read as a conservative lower bound.

6.4 Ground Deformation

Table 6.4 compares mean InSAR coherence at the two confirmed instability zones against the stable baseline.

Table 6.4: Mean InSAR coherence at confirmed instability zones versus the stable baseline.

Location	Mean coherence
Zarrouk zone (confirmed instability)	0.44
Southwest zone (confirmed instability)	0.47
Stable / negative baseline	0.51

Coherence drops modestly at both confirmed zones relative to the stable baseline, by roughly 0.04 to 0.07. That is the direction expected for ground disturbance, though not a dramatic, standalone signal. A few factors likely explain why: the coherence pixels are coarse relative to the small, fragmented bare-soil patches, a single interferometric pair was used rather than a full multi-pair time series, and vegetation or land-cover differences unrelated to ground movement can confound the result. Throughout this work, the coherence result is treated as corroborating, low-weight evidence rather than a primary driver of the final risk score.

6.5 Exposure and Vulnerability

Figure 6.8 shows the two vulnerability proxies side by side: proximity to a confirmed instability zone on the left, and building footprint size on the right.

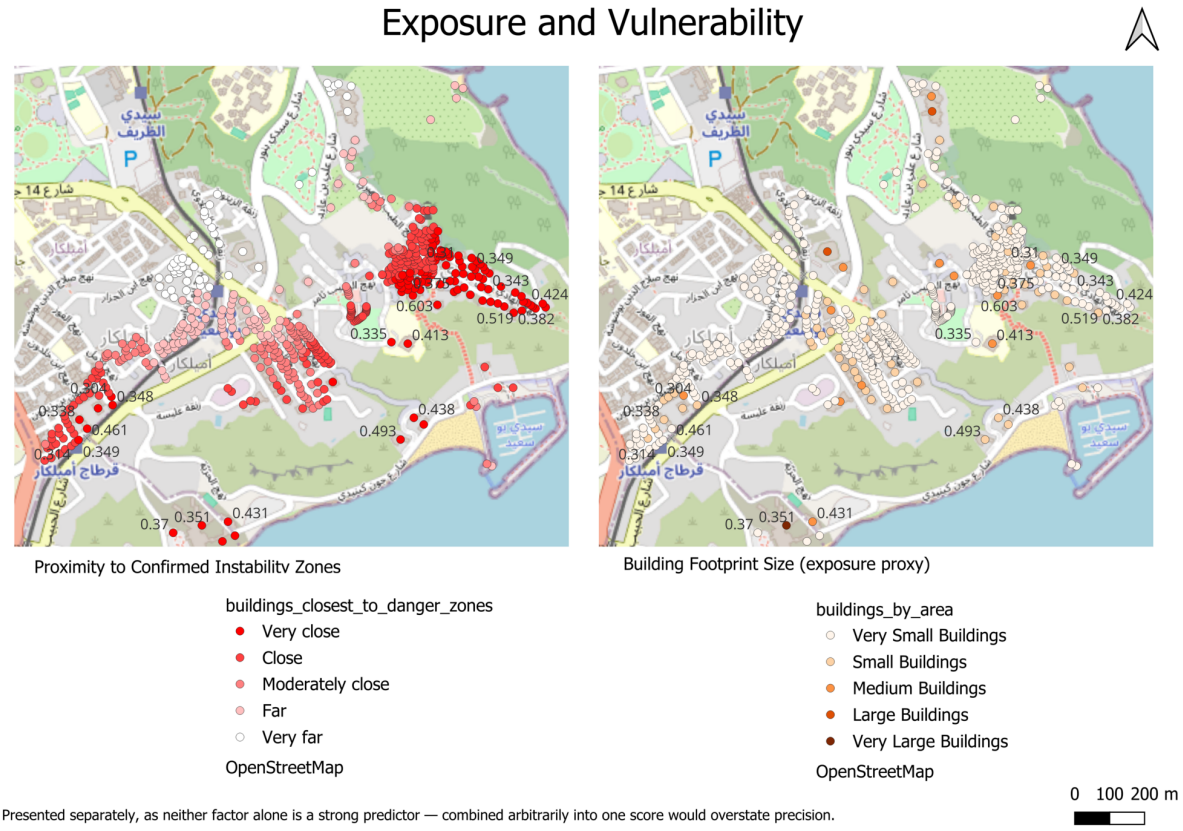


Figure 6.8: Exposure and vulnerability proxies (left: proximity to confirmed instability zones; right: building footprint size).

These two layers, together with the susceptibility raster from Section 6.1, feed directly into the final risk score presented next.

Chapter 7

Result: Final Risk Map

Figure 7.1 brings everything together: the per-building risk score, computed with Equation 5.2, shown over the susceptibility raster for context.

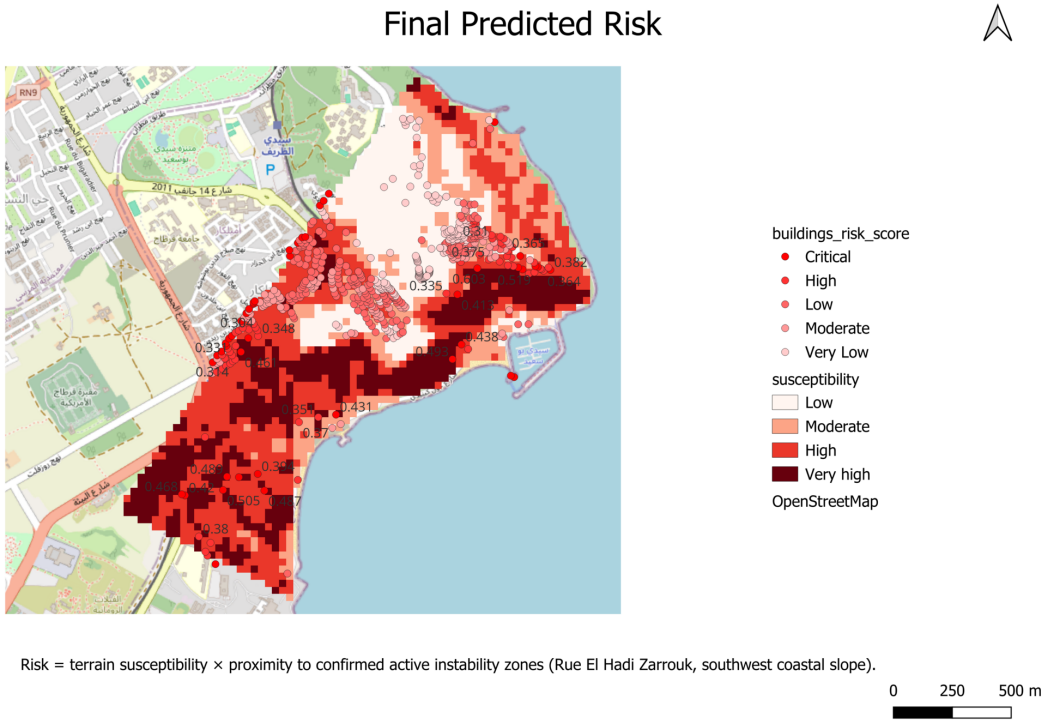


Figure 7.1: Final predicted per-building risk map, combining susceptibility (Equation 5.2) with proximity-based exposure, overlaid on the susceptibility raster.

Table 7.1 defines the five risk categories used in the map's legend, based on Jenks natural breaks of the risk score.

Table 7.1: Risk categories used in the final map legend, based on Jenks natural breaks of `risk_score`.

Category	**risk_score range**	**Description**
Very Low	0.000–0.069	Stable terrain, far from confirmed instability
Low	0.069–0.148	Low hazard or moderate distance from instability
Moderate	0.148–0.246	Elevated terrain hazard
High	0.246–0.382	Hazardous terrain, near active instability
Critical	0.382–0.603	Hazardous terrain and confirmed nearby instability

This map and its category breakdown are the project's main deliverable, and the next chapter checks how well they line up with what actually happened during Storm Harry.

Chapter 8

Validation

Buildings inside the original documented closure corridor were compared against the broader building set. Buildings in and around the Rue El Hadi Zarrouk and Avenue John Kennedy intersection consistently stood out with elevated risk scores, several above 0.4 to 0.6, including the single highest-scoring building in the entire dataset at roughly 0.603, while most buildings elsewhere on the hillside scored well under 0.2. This is the project's central validation result: **the model's highest-risk predictions cluster in the area that was actually closed and partially evacuated during Storm Harry**, even though the model was never told which addresses were evacuated. It learned only from terrain covariates and from bare-soil zones derived independently from satellite imagery. As discussed in Chapter 9, this check supports consistency with the real event but is not a fully independent test.

Chapter 9

Discussion and Limitations

1. **Single-event calibration.** The entire label set traces back to evidence from one storm. That is far more grounded than an inventory-free or arbitrary approach, but the model has not been tested against other, independent triggering events.
2. **Baseline-date asymmetry.** The two confirmed zones rely on pre-event imagery from different dates (2024 for Zarrouk, 2021 for the southwest zone). This was addressed by reporting annualized expansion rates, but it remains worth flagging.
3. **CHIRPS underestimates rainfall magnitude.** The measured 115 to 150 mm totals likely understate the true, more intense localized rainfall. Timing and threshold calibration hold up, but the absolute figures should not be over-interpreted.
4. **Single-pair InSAR, not full SBAS.** This was a deliberate, stated scoping decision given the project timeframe. The resulting coherence signal is real but weak, and is read as corroborating rather than primary evidence.
5. **Elevation's role is partly a spatial-clustering artifact.** This is addressed through the sensitivity check in Section 6.1, but it illustrates a general risk in small-sample susceptibility modeling.
6. **Weak vulnerability proxies.** Footprint area and distance to instability are reasonable, available stand-ins, but they are no substitute for real occupancy, construction-quality, or property-value data, none of which were accessible for this project.
7. **Coarse rainfall resolution.** The roughly 5.5 km CHIRPS pixel size is a genuine scale mismatch against the fine-grained terrain analysis, though this is expected given that rainfall is inherently a regional trigger.
8. **Partial circularity of the validation.** The proximity term in Equation 5.2 is computed from the confirmed instability zones, and the Zarrouk zone lies inside the documented closure corridor. High scores near the corridor are therefore partly built in, so the validation shows consistency with the real event rather than fully independent confirmation. A stronger test would use damage or evacuation records from outside the labeled zones.

Chapter 10

Conclusion and Future Work

This project built a complete, evidence-grounded GeoAI pipeline for per-building landslide risk mapping on the Sidi Bou Said hillside, based throughout on a real, documented triggering event rather than a generic case study. With no formal regional landslide inventory to draw on, a defensible label set was constructed from first principles using multi-source remote sensing evidence, and a hybrid statistical and machine-learning approach to static susceptibility was cross-validated both internally, through an elevation sensitivity check, and externally, through a separate Frequency Ratio analysis. The resulting per-building risk map was then compared against the real evacuation and closure corridor, and the high-risk predictions landed where they should: concentrated in the area that was actually affected.

Several extensions could strengthen this pipeline going forward. Incorporating multiple independent triggering events would reduce the single-event calibration risk. Processing a full multi-temporal SBAS or PSInSAR deformation time series would give a richer view of ground movement. Acquiring real building vulnerability data, such as construction type, occupancy, and age, would replace the current proxies with genuine exposure measures. And extending the AOI to cover the full village and surrounding hillside, rather than just the immediate closure corridor and the two confirmed instability zones, would broaden the map's practical reach.